CHATBOT APPLICATION FOR INDUSTRIAL LAWS

Suttidech Jittawisuttikul
Computer Engineering
of Mae Fah Luang University
Thasud, Muang, Chiang Rai
suttidech_j@hotmail.com

Kanchit Pamanee
INNWISE Co., LTD
Bangkok, Thailand

Roungsan Chaisricharoen
Computer and Communication Engineering for Capacity Building Research Center, School of Information Technology, Mae Fah Luang University, Chiang Rai, Thailand 57100

***Abstract*— In our present time, Speech and textual information play a more important role in human communication than counting face-to-face exchanges. An article in the "The New York Times" published that adults today spend more than eight hours a day looking at computer screens or mobiles. This is done through web applications such as WhatsApp, Facebook, and Twitter among others in the form of speech and text conversations. However, concerning study material and teaching aids, there is very little relevant media available.**

In this research paper, we built a chatbot in the education domain. The proposed chatbot assists in answering questions provided by the users as an additional expedient way to study for students in modern times. The chatbot that has been developed focuses on Industrial law. Because most students in Thailand have difficulty using English, the created chatbot is in the Thai language.

My chatbot uses NLP to build a chatbot system that can understand and respond to questions asked by users. To develop the chatbot system, I have employed an ensemble learning method, the Neuron Network in the presence of extracted features from our prepared dataset. The chatbot has a successful response in answering users' questions with about an eighty seven percent accuracy.

Keywords*—*Chatbots, NLP, Neural network

I. INTRODUCTION

Chatbots, also known as chat robots, are software agents that simulate human conversations via text or voice. The main goal of chatbots is to be like clever humans. It knows the answers to most questions and answers it like a human would. In this case, it created many kinds of machine learning and deep learning methods. It is to be used to develop software that can think much like humans.

Currently, there is very little information available concerning chatbots and industrial law. Therefore, my interest is to develop a chatbot system that is able to answer questions about industrial laws. Reading legal books is time consuming and legal information may not be easily accessible to the general public. Legal books are also very expensive, even legal books written for the general public.

Current law, including industrial law, is often compartmentalized meaning that there are many sections and sub-sections referring back to the main title. This makes it very confusing for a user to find relevant search information. A user would have to know what section their pertinent information is located.

This makes the use of knowledge obtained from typical search ineffective. For example, if you want to search for laws regarding “Industrial Steam Boilers in factories” We may have to spend time gathering various legal texts from the Constitution, Acts, Royal Decrees, and Ministerial Regulations, including announcements, regulations, and requirements. This could take an extensive amount of time and effort to obtain comprehensive information.

We want to develop a superb system to be able to gather scattered industrial legal knowledge and put it in one place. This system will provide concise answers to those who want answers about industrial law. Users can ask questions and be provided the answers through an Ai Chatbot system, which is an online application that is easily accessible. Chatbots are convenient to use and

replace the search functions in Browsers, searching by reading legal books, hiring an attorney or studying from other online media.

In this project, four classifier methods are used to create four chatbots. In the final comparison analysis, the Neuron Network was selected as the best method to assist the user in their quest for relevant answers pertaining to industrial law.

II. LITERATURE REVIEW

Using machine learning algorithms to develop chatbots eliminates the need to define and code new pattern-matching rules. This allows chatbots to be more flexible and no longer dependent on domain-specific knowledge. As noted, AI models can be divided into retrieval-based models and general-purpose models. An extraction model called the fetch-based model is designed to provide a text-based dataset. The algorithm will be able to retrieve the necessary information based on user input. The algorithms used are usually shallow learning algorithms. However, there are cases of retrieval models that use rules-based algorithms and deep learning models for a predefined set of possible answers.

A chatbot processes the user's search query and based on this input will choose one of the answers contained in the set of the knowledge base for this type of model that was built from a database of question-and-answer pairs. A conversation index is generated from this database to predict all possible answers according to the prompts and choose by the accuracy rate of the prediction function. When a user provides input to a chatbot, the chatbot will treat the input as a query. A fetching model similar to that used for web browsing is used to match user input to similar data in the chat index.

The results returned to the user are therefore answers that are matched to the selected questions listed in the Discussion Index. The main advantage of this model is that it ensures the quality of the answers because they are not automatically generated. These models have gained immense popularity with the advent of Web 2.0 and the rise of text-based information that can be retrieved on social media platforms, forums, and chats. The approach to building the necessary knowledge base is costly, time-consuming, and tedious. It also means that matching user input with correct answers becomes more challenging. Training the system to choose the correct answer takes a lot of time and resources.

III. METHODOLOGY

I have gathered knowledge about industrial laws then converted the dataset from the text file to the dataset in a programming language format, such as the JSON format. After the question-answer pair dataset format has been changed, a system was developed to use in preparing the data in the form of an input dataset for the system by the following steps. Examples of words are in English so that the examples may be widely understood.

Step 1. Collecting Words

This step is word processing to read all of the words from datasets. We need to collect all of the words to create a set of features, this is a standard requirement of input data for the deep learning methods we used

Questions	Answers
Q1: What is a cat?	A1: The cat is similar in anatomy to the other felid species: it has a strong flexible body, quick reflexes, sharp teeth, and retractable claws adapted to killing small prey like mice and rats.
Q2: What is a dog?	A2: The dog (*Canis familiaris* or *Canis lupus familiaris*) is a domesticated descendant of the wolf. Also called the domestic dog, it is derived from extinct Pleistocene wolves, and the modern wolf is the dog's nearest living relative.
Q3: What is a Tiger?	A3: The horse (Equus ferus caballus) is a domesticated, one-toed, hoofed mammal. It belongs to the taxonomic family Equidae and is one of two extant subspecies of Equus ferus. The horse has evolved over the past 45 to 55 million years from a small multi-toed creature, close to Eohippus, into the large, single-toed animal of today.

Table 1. The example question-and-answer pair

Step 2. Word filters and sorting

In this first optimization technique, duplicate words are sorted and filtered out from the input dataset.

For example:

<u>Input</u>: “What is a cat”, “What is a dog”, “What is a tiger”

<u>Result</u>: “a” “cat” “dog” “is” “tiger” “what”

Step 3. Stemming words

This step is the second optimization technique used to group the same meaning of words and convert them to one form. The word "get" has past particles such as, "got", "gotten" and "getting." This group of words basically represent the same meaning and it is necessary to convert all the words in this group to the root "get" to reduce the size of the dataset. This is a powerful tool with no negative effects.

Step 4. Transforming the stemmed words into the features table

This step is contingent on the dataset. With more unique words, we have more features for deep learning methods to process. All the words must be sorted and then a table is created called a features table that contain all the word indexes by sorting the words in ascending order.

	a	cat	dog	is	tiger	what
0						
1						
2						
…						
n						

Table 2. The features table

Step 5. Transform the "question sentence" to a features table

This step is important and takes some time to complete. Completion times will depend on the hardware used because all the steps above must be accomplished for every question sentence. Each word in the "question sentence" must be indexed in the same position as the features table.

for example:

We have the words of the features table shown in "Table2." When we have the question "What is a cat?" This question must be transformed to "a cat is what" and inserted in the features table below:

	a	cat	dog	is	tiger	what
a cat is what	1	1	0	1	0	1

Table 3. The mapping result between questions and features table.

If there is an "a" in the sentence, a 1 is inserted in the column index of the word "a" in the features table, replete until the end of words in the transformed sentence. The result of the question "What is a cat" is listed as [1, 1, 0, 1, 0, 1] and inserted in the training data table (*Table 4*.) In this case, each word represents each feature, therefore, the input of "What is a tiger?" is listed as [1, 0, 0, 1, 1, 1.] This process is repeated for all of the question sentences in the dataset before sending to the training process.

Step 6. Training the Chatbots

In this step, we train all of the chatbots that we created by using a "Decision Tree classifier", "Naïve Bayes classifier","KNeighbors Classifier", and a "Neural Network (Multilayer Perceptron classifier)" with the same dataset and preserve the results of the training.

	a	Cat	dog	Is	tiger	what	answer
Q1	1	1	0	1	0	1	A1
Q2	1	0	1	1	0	1	A2
Q3	1	0	0	1	1	1	A3

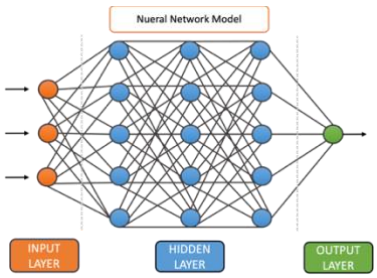
Table 4. The training data (the input of the training process)

Step 7. Testing Prediction

In this step, a simulated system for automatic testing was created to maximize the test procedures of the chatbots to 1000, 10,000, and 50,000 runtimes to collect prediction results and send them to the reporting system

Step 8. Reporting

This is the final step of the training process and will prove which type of classifier will return the best appropriate answer for our dataset. Based on the chatbot testing procedures, the Neuron Network model provided the most accurate results.

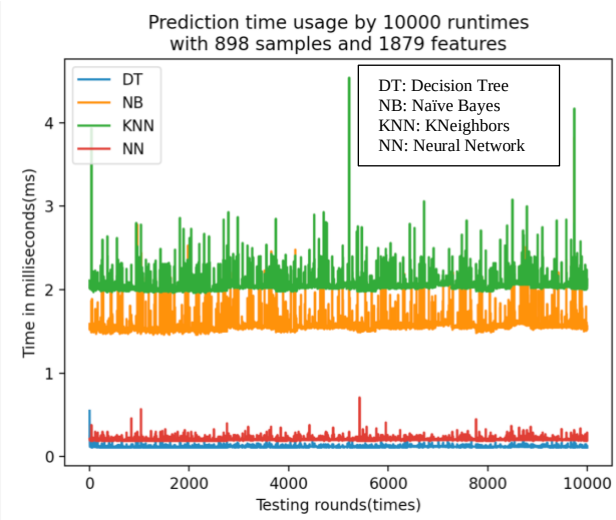


Flg1. Neuron Network model

Proposed System

A. Difference Chatbot

We developed four different chatbots based on a Decision Tree classifier, Naïve Bayes classifier,
KNeighbors Classifier, and a Neural Network classifier for comparison to find the appropriate classifier with our input datasets.



Flg2. The prediction time usage comparison report generates.

Methods	Training time(ms)	Train Accuracy(%)	Test time	Test Accuracy
DT:Decision Tree Classifier	92.233	77.222	0.12	95.47
NB:Gaussian Naive Bayes	15.054	57.778	1.562	91.52
KNN:KNeighbors Classifier	1.517	62.222	2.043	74.27
NN:Neural Network Classifier	3779.586	80.556	0.204	96.07

Flg3. The testing accuracy result comparison report table is generated by Python.

B. Data Preparation system

Pre-processing is applied to the input text to standardize the input as per the system's requirement based on the programming syntax and Input optimization.

C. Prediction and Simulation Results Collection and Reporting

We have developed a simulation system to retain the results from the predictions by each chatbot classifier and generate the results through charts and figures

D. Artificial Intelligence Chatbots

The AI model chatbots are based on a machine-learning algorithm that allows them to learn from the existing database of human conversations. To do so, they must be trained through machine learning algorithms that can be trained. There is no longer any need to define and code the system, it just needs an input dataset in the required form for the user to provide questions and answers.

IV. Future Work

The experience of having created chatbots for use in answering general knowledge questions that can be retrained over time has inspired me to make chatbots easy to use and responsive to real-world demands. Further development in these areas will add greater value and accuracy to user's questions.

1) *Increase the accuracy of answering questions.*
2) *Increase the ability to receive conversations by speaking. Simulate having a conversation with a real person.*
3) *Increase multilanguage abilities.*

V. Conclusion

Chatbots are an effective, efficient, quick and accurate way for users to seek answers to their questions and access data concerning complex Industrial Laws.

The purpose of a chatbot system is to simulate human conversations. Natural language processing is used in conjunction with Python to develop a chatbot. Simple but automated this can be easily trained and applied in any field depending on the dataset used to teach the chatbot. It is also possible to teach chatbots to learn new knowledge or modify existing knowledge to keep it accurate and current. The trainer or administrator can train and update the chatbot's knowledge base at any time enabling the chatbot to have the most pertinent data.

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